



Operating Instructions

Diesel Engine
20V4000G44F
20V4000G44LF
20V4000G94F
20V4000G94LF

MS15058/09E

Engine model	Power	Frequency	Application group
20V4000G44F	140 kW/cyl.	50 Hz	3B, continuous service, variable load, ICXN
20V4000G44F	140 kW/cyl.	50 Hz	3F, heavy duty for DCP, unrestricted, ICXN
20V4000G44LF	150 kW/cyl.	50 Hz	3E, emergency service, ICXN
20V4000G44LF	150 kW/cyl.	50 Hz	3F, heavy duty for DCP, unrestricted, ICXN
20V4000G94F	154 kW/cyl.	50 Hz	3D, emergency standby power, IFN
20V4000G94LF	165 kW/cyl.	50 Hz	3D, emergency standby power, IFN
Load factor:	$\leq 75\%$ (3B) $\leq 85\%$ (3D, 3E) $\leq 100\%$ (3F)		
Operating hours:	No limit for operating hours (3B, 3F) 500 h per year or for the duration of an emergency (3D, 3E)		
Time between major over-haul:	30,000 h / 24,000 h (3B) 8,000 h or 18 years (3D, 3E, 3F)		

Table 1: Applicability

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California Proposition 65

This warning applies only to the State of California, USA, in compliance with Title 27 California Code of Regulations, Article 6, Clear and Reasonable Warnings.

 **WARNING:** Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to www.P65warnings.ca.gov/diesel.

(→ www.P65warnings.ca.gov/diesel)

Table of Contents

1	Preface		6.2	Product data 20V4000G44LF	69
1.1	Preface	8	6.3	Product data 20V4000G94F	74
2	Safety		6.4	Product data 20V4000G94LF	78
2.1	Important provisions for all products	9	6.5	Firing order	82
2.2	Important requirements for Power Generation application products	10	6.6	Engine- Main dimensions	83
2.3	Intended use of gendrive engines	12	7	Operation	
2.4	Personnel and organizational requirements	13	7.1	Preparations for putting into operation after extended out-of-service periods (>3 months)	84
2.5	Safety regulations for commissioning and operation	14	7.2	Preparations for putting into operation after scheduled out-of-service-period	85
2.6	Safety regulations for assembly, maintenance, and repair work	16	7.3	Starting the engine in manual mode (test run)	86
2.7	Fire and environmental protection, indirect materials	20	7.4	Engine - Quick-start function	87
2.8	Standards for warning messages in the text and highlighted information	23	7.5	Re-starting the engine following an automatic safety shutdown	88
3	Transport		7.6	Safety system - Override	89
3.1	Transport	24	7.7	Operational monitoring	90
3.2	Lifting requirements	25	7.8	Emergency engine start (Override mode)	91
3.3	Transport locking device for crankshaft	26	7.9	Engine - Stopping in manual mode (test run)	92
3.4	Engine mounting transportation lock	29	7.10	Emergency engine stop	94
4	General Information		7.11	After stopping the engine - Engine remains ready for operation	95
4.1	Engine side and cylinder designations	31	7.12	After shutting down engine - Putting engine out of operation	96
4.2	Data acquisition and data transmission	32	7.13	Plant - Cleaning	97
5	Functional Description		8	Troubleshooting	
5.1	Engine layout	33	8.1	Troubleshooting	99
5.2	Crankcase with oil pan	35	8.2	Fault messages of the Engine Control Unit (ECU 9) for Series 4000 engines, application Genset	102
5.3	Gear train	37	8.3	Fault conditions of the alarms 1801 to 1840	143
5.4	Running gear	38	9	Maintenance	
5.5	Cylinder head with injector	40	9.1	Maintenance task reference table [QL1]	146
5.6	Valve gear	42	10	Task Description	
5.7	Fuel system with common-rail injection	44	10.1	Engine	148
5.8	Charge-air and exhaust system	47	10.1.1	Barring engine manually	148
5.9	Lube oil system	49	10.1.2	Engine - Cranking on starting system	150
5.10	Cooling system	51	10.1.3	Engine - Test run	151
5.11	Fuel cooler	54	10.2	Cylinder Liner	152
5.12	Engine management and monitoring	56	10.2.1	Cylinder liner - Endoscopic inspection	152
5.13	Sensors and actuators - Overview	59	10.2.2	Cylinder liner - Instructions and comments on endoscopic and visual examination	154
6	Technical Data				
6.1	Product data 20V4000G44F	64			

10.3	Crankcase Breather	156	10.13.2	Centrifugal oil filter - Cleaning and filter sleeve replacement	216
10.3.1	Crankcase breather - Oil mist separator replacement	156	10.14	Coolant Circuit, General, High-Temperature Circuit	219
10.3.2	Crankcase breather - Filter element replacement	157	10.14.1	Engine coolant - Level check	219
10.4	Valve Drive	159	10.14.2	Engine coolant - Change	221
10.4.1	Valve protrusion - Measurement	159	10.14.3	Engine coolant - Draining	222
10.4.2	Valve gear - Lubrication	163	10.14.4	Engine coolant - Filling	224
10.4.3	Valve clearance - Check and adjustment	164	10.14.5	Engine coolant pump - Relief bore check	228
10.4.4	Cylinder head cover - Removal and installation	167	10.14.6	Engine coolant - Sample extraction and analysis	229
10.5	Injection Pump / HP Pump	168	10.14.7	Preheater - Overview	231
10.5.1	HP pump - Filling with engine oil	168	10.14.8	Preheater - Overhaul	232
10.6	Injector	171	10.14.9	Preheater - Function and leak check	233
10.6.1	Injector - Replacement	171	10.14.10	Preheater - Heating element and housing check	234
10.6.2	Injector - Removal and installation	172	10.14.11	Preheater - Solenoid switch replacement	237
10.7	Fuel System	178	10.14.12	Preheater thermostats - Replacement	239
10.7.1	Fuel system - Venting	178	10.14.13	Preheater - Replacement	242
10.8	Fuel Filter	180	10.14.14	Preheater - Removal	243
10.8.1	Fuel filter - Replacement	180	10.14.15	Preheater - Installation	244
10.8.2	Additional fuel filter - Replacement	182	10.14.16	Coolant expansion tank - Valve cover replacement	245
10.8.3	Fuel prefilter - Pressure ratios	183	10.15	Low-Temperature Circuit	247
10.8.4	Fuel prefilter - Draining	184	10.15.1	Charge-air coolant level - Check	247
10.8.5	Fuel prefilter (non-switchable) - Draining	185	10.15.2	Charge-air coolant - Change	249
10.8.6	Fuel prefilter (switchable) - Draining	188	10.15.3	Charge-air coolant - Draining	250
10.8.7	Fuel prefilter (switchable) - Low pressure indicator check	191	10.15.4	Charge-air coolant - Filling	252
10.8.8	Fuel prefilter - Filter element replacement	192	10.15.5	Charge-air coolant pump - Relief bore check	256
10.8.9	Fuel prefilter (non-switchable) - Filter element replacement	193	10.16	Battery-Charging Generator	257
10.8.10	Fuel prefilter (switchable) - Filter element replacement	195	10.16.1	Battery-charging generator - Cleaning	257
10.8.11	Fuel prefilter - Venting	198	10.16.2	Drive belt - Condition check	258
10.9	Charge-Air Cooling	200	10.16.3	Battery-charging generator - Drive belt tension adjustment	259
10.9.1	Intercooler - Checking condensate drain for coolant leak and obstruction	200	10.16.4	Battery-charging generator drive - Drive belt replacement	260
10.10	Air Filter	201	10.17	Storage	261
10.10.1	Air filter - Check	201	10.17.1	Engine mounting - Checking resilient element	261
10.10.2	Air filter - Replacement	202	10.17.2	Engine mounting - Checking securing screws for firm seating	262
10.10.3	Air filter - Removal and installation	203	10.17.3	Engine mount - Measuring height of rubber element	264
10.10.4	Heavy duty air filter - Filter insert and filter element removal and installation	204	10.17.4	Generator mounting - Check	265
10.11	Air Intake	206	10.17.5	Generator mounting - Securing screw firm seating check	266
10.11.1	Service indicator - Signal ring position check	206	10.17.6	Generator mount - Measuring height of rubber element	267
10.12	Lube Oil System, Lube Oil Circuit	207	10.18	Radiator	268
10.12.1	Engine oil - Change	207	10.18.1	Fuel cooler - External contamination and leak-tightness check	268
10.12.2	Engine oil - Level check	210	10.18.2	Fuel cooler - Cleaning	269
10.12.3	Engine oil - Sample extraction and analysis	211	10.19	Engine Governor	271
10.12.4	Oil priming pump - Visual inspection	213	10.19.1	Engine governor ECU 9 - Removal and installation	271
10.13	Oil Filtration / Cooling	214			
10.13.1	Engine oil filter - Replacement	214			

10.19.2	Engine governor plug connections - Check	272	11.2	Conversion tables	280
10.20	Wiring (General) for Engine/Gearbox/Unit	273	11.3	Contact person/Service partner	284
10.20.1	Engine cabling - Check	273			
10.21	Accessories for (Electronic) Engine Governor / Control System	274	12	Appendix B	
10.21.1	Resetting CDC parameter and entering IIG with DiaSys®	274	12.1	Standard Tools and Special Tools	285
10.21.2	Engine governor and connectors - Cleaning	275	12.2	Index	294
11	Appendix A				
11.1	Abbreviations	276			

1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:<http://www.mtu-solutions.com>

2.2 Important requirements for Power Generation application products

Project-specific operational environment

During integration and use of the diesel engine, all currently valid and relevant standards and guidelines on safety and specified operation must be implemented and observed. This applies to persons responsible for:

- The installation of the engine
- The design of the engine-generator set
- The installation of the engine-generator set

The relevant standards are, in particular:

- DIN EN 1679-1
- EN ISO 8528-13

Residual hazards

In the event of a defect, the diesel engine can unexpectedly have insufficient or no drive power at all. We therefore recommend a redundant design of the engine-generator set system in safety-critical applications.

Protection against hot surfaces

Combustion engines and their auxiliary components have hot surfaces (e.g. exhaust system, exhaust turbochargers, exhaust aftertreatment system, engine oil system, cooling system), which can cause injury to persons who approach or touch them (danger of burns).

- In operation
- During the cooling-down phase after operation

The plant integrator and operator must assess, based on spatial or other plant conditions, whether the danger potential usually associated with the state of technology is exceeded, e.g.:

- Particularly narrow maintenance spaces
- Particularly exposed position of hot surfaces

In such cases, the plant integrator or operator must equip the plant with the necessary and suitable safety devices to reliably exclude the possibility of personal injury, e.g.:

- Heat shields
- Insulation

The operator must ensure that the plant is not operated without these protective devices. Operating and maintenance personnel must be informed about the danger of hot surfaces and the prohibition of operation without the protective devices.

Noise

Noise protection measures that comply with statutory requirements must be implemented by the plant integrator and operator.

Chemical resistance

Only use hoses, pipes, fittings, etc. that can tolerate the fluids and lubricants in accordance with the Fluids and Lubricants Specifications. Only use options that are resistant to the fluids and lubricants in accordance with the Fluids and Lubricants Specifications.

Electrical devices

The IP protection class of the electrical devices required by the manufacturer must be observed.

Hoses

For vibration decoupling, where possible only use hoses or rubber sleeves. Install devices in low-vibration areas and not on the engine. Hoses must be installed such that they do not chafe and are not under tension.